



ONE SEMESTER PROJECT

USER MANUAL DOCUMENT

24/06590

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BDS 3106

BACHELOR OF SCIENCE IN DATA SCIENCE

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PROJECT TITLE : SMART BANKLYTICS

An Intelligent Data – Driven Bank Management and Customer Analytics System

Table of Contents

1.0 INTRODUCTION	4
1.1 Purpose	4
1.2 System Overview	4
1.3 Intended Audience	4
2.0 SYSTEM REQUIREMENTS.....	4
3.0 ACCESSING THE SYSTEM	5
3.1 Opening the System	5
4.0 USER GUIDE – CUSTOMER	5
4.1 Logging In	5
4.2 Customer Dashboard	5
4.3 Depositing Funds.....	5
4.4 Withdrawing Funds	6
4.5 Transferring Funds.....	6
4.6 Viewing Transaction History.....	7
4.7 Downloading a PDF Bank Statement	7
4.8 Fraud Alerts	8
4.8.1 Appealing a Fraud Alert	8
4.9 Password Recovery	9
4.10 Logging Out	9
5. USER GUIDE – ADMINISTRATOR	9
5.1 Logging In as Administrator	9
5.2 Admin Dashboard Overview	10
5.3 Creating a New User Account	10
5.4 Viewing All Users.....	11
5.5 Managing a User Account	11
5.5.1 Depositing Funds into a User Account.....	11
5.5.2 Suspending or Activating a User Account	12
5.6 Reviewing Fraud Alerts	12

5.7 Viewing the Analytics Dashboard	12
6.0 SYSTEM MESSAGES REFERENCE.....	13
7.0 FRAUD DETECTION REFERENCE	14
8.0 CONCLUSION	14

1.0 INTRODUCTION

1.1 Purpose

This User Manual provides step-by-step guidance on how to use the Smart Banklytics system. It is intended for two categories of users: customers who perform banking transactions, and administrators who manage the system. This manual explains how to navigate the system, perform operations, and understand system responses.

1.2 System Overview

Smart Banklytics is a web-based intelligent banking system that allows customers to manage their bank accounts and perform financial transactions, while administrators monitor system activity, manage users, review fraud alerts, and analyze banking performance through an analytics dashboard.

1.3 Intended Audience

- **Customers** – Bank clients who use the system to manage their accounts and perform transactions.
- **Administrators** – Authorized bank staff responsible for managing users and monitoring system operations.

2.0 SYSTEM REQUIREMENTS

To access Smart Banklytics, the following are required:

Requirement Detail

Device	Computer, laptop, or internet-enabled device
Browser	Google Chrome, Mozilla Firefox, or any modern web browser
Internet	Stable internet or local network connection
Credentials	Valid username and password provided during registration

3.0 ACCESSING THE SYSTEM

3.1 Opening the System

Open your web browser and navigate to the Smart Banklytics application URL or local address (e.g., <http://127.0.0.1:5000>). The Login page will be displayed.

4.0 USER GUIDE – CUSTOMER

4.1 Logging In

1. On the Login page, enter your **Username** in the username field.
2. Enter your **Password** in the password field.
3. Click the **Login** button.
4. If your credentials are correct, you will be redirected to your **Customer Dashboard**.
5. If your credentials are incorrect, an error message will be displayed: *"Invalid username or password!"*

Note: If your account has been suspended, contact your bank administrator.

4.2 Customer Dashboard

Once logged in, your dashboard displays the following information:

- Your full name and account number
 - Your current account balance
 - Your account type (Savings / Current / Student)
 - Your recent transaction history
 - Any fraud alerts linked to your account
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4.3 Depositing Funds

Note: Deposits are processed by the administrator on behalf of customers at the bank counter.

The administrator accesses your account via the Admin Panel and deposits funds directly into your account. Your balance will reflect the deposit immediately, and the transaction will appear in your dashboard transaction history.

4.4 Withdrawing Funds

1. From your dashboard, navigate to the **Withdraw** page.
2. Enter the **Amount** you wish to withdraw.
3. Click the **Withdraw** button.
4. If the withdrawal is successful, your balance will be updated and a confirmation message will be displayed: *"Withdrew KES [amount] successfully!"*

Possible Error Messages:

Message	Cause
Amount must be greater than zero	Entered amount is zero or negative
Insufficient balance	Withdrawal amount exceeds current balance
Your account is suspended	Account has been suspended by the system or administrator
Suspicious activity detected. Account suspended	Transaction triggered a fraud detection rule

4.5 Transferring Funds

1. Navigate to the **Transfer** page from your dashboard.
2. Enter the **Recipient's Account Number** in the designated field.
3. Enter the **Amount** you wish to transfer.
4. Click the **Transfer** button.
5. If successful, a confirmation message will be displayed: *"Transferred KES [amount] to [Recipient Name]!"*

Possible Error Messages:

Message	Cause
Amount must be greater than zero	Entered amount is zero or negative
Sender and receiver must be different	You entered your own account number as the recipient
Invalid recipient account	The recipient account number does not exist
Insufficient funds	Transfer amount exceeds your current balance
Your account is suspended	Account has been suspended
Suspicious activity detected. Account suspended	Transaction triggered a fraud detection rule

4.6 Viewing Transaction History

Your full transaction history is displayed on your **Customer Dashboard** under the transaction history section. Each transaction entry shows:

- Transaction type (Deposit, Withdrawal, Transfer In, Transfer Out)
 - Amount in KES
 - Date and time of the transaction
 - Source account (for transfers)
 - Destination account (for transfers)
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4.7 Downloading a PDF Bank Statement

1. On your dashboard, locate the **Download Statement** option.
2. Select a **Filter Option**:
 - **Past 30 Days** – Transactions from the last 30 days
 - **Past 14 Days** – Transactions from the last 14 days
 - **Custom Range** – Enter a specific start and end date
3. Optionally, filter by **Transaction Type** (Deposit, Withdrawal, Transfer In, Transfer Out).

4. Click the **Download** button.
5. A PDF file named [username]_transactions.pdf will be downloaded to your device.

The PDF statement includes:

- Bank logo
- Your full name and account number
- Current balance
- Selected date range
- Full transaction table with date, type, amount, and transfer details

4.8 Fraud Alerts

If the system detects suspicious activity on your account, a fraud alert will be created and your account may be suspended. You can view your fraud alerts on your **Customer Dashboard**.

Understanding Alert Statuses:

Status	Meaning
Pending	Alert is awaiting admin review
Reviewed	Admin has reviewed the alert
Investigating	Admin is investigating the transaction
Frozen	Account has been frozen by the admin
Appealed	You have submitted an appeal
Legitimate	Transaction was confirmed as genuine

4.8.1 Appealing a Fraud Alert

If you believe a fraud alert on your account is incorrect, you can submit an appeal:

1. On your dashboard, locate the fraud alert you wish to appeal.
2. Click the **Appeal** button next to the alert.

3. A confirmation message will be displayed: *"Your appeal has been submitted. Admin will review it shortly."*
4. The alert status will be updated to **Appealed** and the administrator will be notified.

Note: You can only appeal alerts with a status of Pending, Investigating, or Frozen.

4.9 Password Recovery

If you have forgotten your password:

1. On the Login page, click **Forgot Password**.
2. Enter your **Username, Account Number, and National ID Number**.
3. Enter your **New Password** and confirm it in the **Confirm Password** field.
4. Click **Reset Password**.
5. If your details are verified, a success message will be displayed and you will be redirected to login with your new password.

Possible Error Messages:

Message	Cause
Passwords do not match	New password and confirm password fields are different
Details do not match any account	The entered credentials do not match any record

4.10 Logging Out

To log out of the system, click the **Logout** button available on your dashboard. You will be redirected to the Login page and your session will be terminated.

5. USER GUIDE – ADMINISTRATOR

5.1 Logging In as Administrator

1. On the Login page, enter your administrator **Username** and **Password**.

2. Click **Login**.
 3. You will be redirected to the **Admin Dashboard**.
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5.2 Admin Dashboard Overview

The Admin Dashboard provides a complete overview of the banking system including:

- **KPI Summary Cards:**
 - Total registered users
 - Total transactions processed
 - Total deposits
 - Total withdrawals
 - Total transfers
 - Total bank balance
 - **Analytics Charts:**
 - Transaction trend chart (deposits, withdrawals, transfers over time)
 - Cumulative bank balance chart
 - Account type distribution chart
 - **Fraud Alert Notifications:** A notification badge displays the number of pending or appealed fraud alerts requiring attention.
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5.3 Creating a New User Account

1. From the Admin Dashboard, navigate to the **Create Account / Signup** page.
2. Fill in the following details:
 - First Name
 - Last Name
 - National ID Number
 - Username

- Password and Confirm Password
 - Account Type (Savings / Current / Student)
3. Click the **Create Account** button.
 4. The system will auto-generate a unique account number for the new user.
 5. A success message will confirm account creation: *"User account created successfully!"*

Possible Error Messages:

Message	Cause
Passwords do not match	Password fields do not match
Username or account number already exists	Duplicate username detected

5.4 Viewing All Users

1. From the Admin Dashboard, navigate to **View Users**.
 2. A full list of all registered users will be displayed showing name, username, account number, account type, balance, role, and status.
 3. Click **Manage** next to any user to open their detailed management page.
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5.5 Managing a User Account

The **Manage User** page allows the administrator to:

- View the user's full account details
- View the user's complete transaction history
- Review fraud alerts linked to the user's account
- Deposit funds directly into the user's account
- Suspend or activate the user's account

5.5.1 Depositing Funds into a User Account

1. On the **Manage User** page, locate the deposit section.
2. Enter the amount to deposit.

3. Click **Deposit**.
4. The user's balance will be updated and the transaction logged.

5.5.2 Suspending or Activating a User Account

1. On the **Manage User** page, click the **Suspend** button to suspend an active account, or the **Activate** button to reactivate a suspended account.
2. The account status will toggle immediately and a confirmation will be displayed.

5.6 Reviewing Fraud Alerts

Fraud alerts are generated automatically when the system detects suspicious activity. Administrators can review and act on each alert from the **Manage User** page.

Available Actions:

Action	Description
Legitimate	Confirms the transaction as genuine. The blocked transaction is processed, and the account is activated.
Investigate	Flags the alert for further investigation. The account remains suspended.
Freeze	Confirms the account as fraudulent. The account is permanently frozen.

Steps to Review a Fraud Alert:

1. Navigate to the user's **Manage User** page.
2. Locate the fraud alert in the alerts section.
3. Select the appropriate action: **Legitimate**, **Investigate**, or **Freeze**.
4. Click the corresponding button.
5. A confirmation message will be displayed with the outcome of the action.

5.7 Viewing the Analytics Dashboard

The analytics dashboard is accessible directly from the Admin Dashboard and displays:

- Live KPI metrics computed from the database

- A transaction trend line chart showing deposits, withdrawals, and transfers over time
- A cumulative balance chart showing the bank's financial growth over time
- An account type distribution chart showing the breakdown of Savings, Current, and Student accounts

The dashboard updates automatically each time the admin page is loaded, reflecting the most current data in the system.

6.0 SYSTEM MESSAGES REFERENCE

Message	Module	Meaning
Invalid username or password	Login	Credentials do not match any record
Your account is suspended	Withdraw / Transfer	Account has been suspended
Insufficient balance	Withdraw	Not enough funds
Insufficient funds	Transfer	Not enough funds for transfer
Suspicious activity detected. Account suspended	All transactions	Fraud rule triggered
User account created successfully	Signup	New account created
Password reset successful	Forgot Password	Password updated
Your appeal has been submitted	Appeal	Appeal recorded successfully
Alert marked legitimate; transaction processed	Fraud Review	Admin confirmed transaction as genuine

7.0 FRAUD DETECTION REFERENCE

The system automatically monitors all transactions using the following rules:

Rule	Trigger	System Response
Large Transaction	Any single transaction exceeds KES 100,000	Account suspended, fraud alert created
Rapid Transactions	3 or more transactions within a 5-minute window	Account suspended, fraud alert created

When either rule is triggered, the transaction is blocked, the account is suspended, and a fraud alert is created for administrator review.

8.0 CONCLUSION

This User Manual provides complete guidance for both customers and administrators on how to use the Smart Banklytics system effectively. Customers can manage their accounts, perform transactions, download statements, and appeal fraud alerts. Administrators can manage users, process deposits, review fraud alerts, and monitor banking performance through the analytics dashboard. For any issues not covered in this manual, contact your system administrator.